

(1) $M'XW, M'XW, M'X'W, S'X'W$ ID 20241063

Problem #1

الهدف من هذا السؤال هو

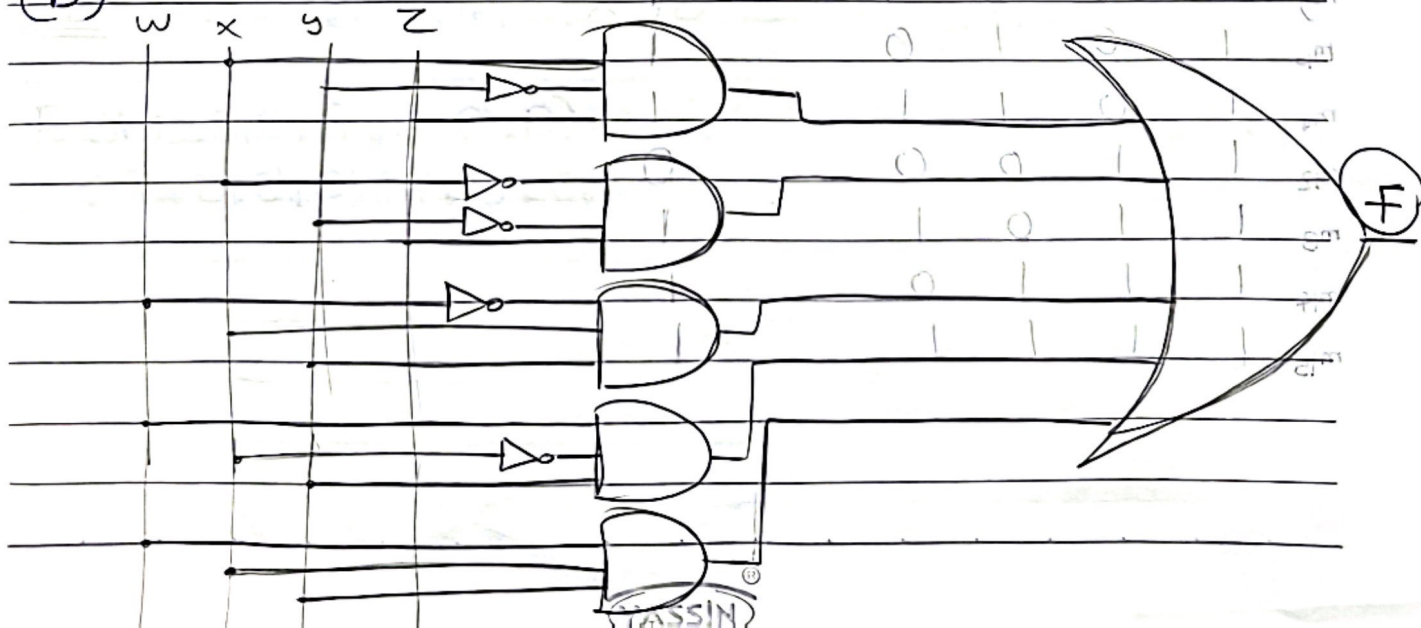
$$F = (xy'z + x'y'z + w'xy + wx'y + wxy) \cdot 15' =$$

$$= (w'xy'z + w'x'y'z + w'xy'z + w'xy'z + w'xy'z) \cdot 15' =$$

$$(a) \quad w \quad x \quad y \quad z \quad (F)$$

m_0	m_1	m_2	m_3	m_4	m_5	m_6	m_7	m_8	m_9	m_{10}	m_{11}	m_{12}	m_{13}	m_{14}	m_{15}
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	1	0	1	0	1	0	1	0	1	0	1	0
	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0
	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0
	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0
	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0
	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0
	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0
	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0

(b)



(C) $F = XY'Z + X'Y'Z + W'XY + WX'Y + WXY$ ID 20241063

$$\begin{aligned}
 &= Y'Z(X+X') + W'XY + WY(X+X') \\
 &= Y'Z \cdot 1 + W'XY + WY \cdot 1 \\
 &= Y'Z + W'XY + WY
 \end{aligned}$$

$$= Y'Z + Y(W'X + W)$$

$$Y(W+W')(W+X) = 1 \cdot W+X = W+X$$

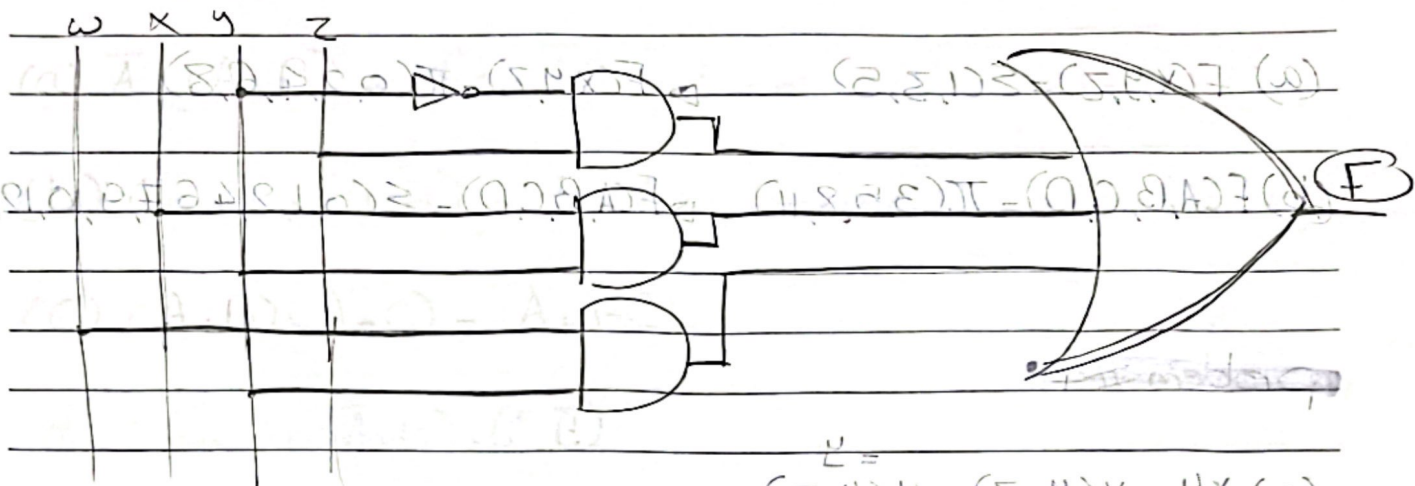
$$Y(W+X)$$

$$= Y'Z + XY + WY$$

(d) $F = \sum(1, 5, 6, 7, 9, 10, 11, 13, 14, 15)$

	w	x	y	z	F
m ₀	0	0	0	0	0
m ₁	0	0	0	1	1
m ₂	0	0	1	0	0
m ₃	0	0	1	1	0
m ₄	0	1	0	0	0
m ₅	0	1	0	1	1
m ₆	0	1	1	0	1
m ₇	0	1	1	1	1
m ₈	1	0	0	0	0
m ₉	1	0	0	1	1
m ₁₀	1	0	1	0	1
m ₁₁	1	0	1	1	1
m ₁₂	1	1	0	0	0
m ₁₃	1	1	0	1	1
m ₁₄	1	1	1	0	1
m ₁₅	1	1	1	1	1

(e) $F = y'z + xy + wy$



$F = (5, 7)z + (5, 6)xy + (6, 7)wx$

Total number of Gates 1 NOT, 3 AND, 1 OR = 5 Gates, $MX =$

in (b) Total number of Gates 5 NOT, 5 AND, 1 OR = 11 Gates

Problem #2

$F(A, B, C, D) = B'D + AD + BD$
 $= D(B' + A + B)1$

expand each term

$\therefore F = D$

$\Rightarrow F = 1$ when $D = 1$

$F = \Sigma(1, 3, 5, 7, 9, 11, 13, 15)$

SOP

$F = A'B'C'D + A'B'CD + AB'C'D + AB'CD + A'BC'D + A'BCD + A'B^*C^*D + AB^*CD$

POS

$F = \Pi(0, 2, 4, 6, 8, 10, 12, 14)$

$F = (A+B+C+D)(A+B+C'+D)(A+B'+C+D)(A+B'+C'+D)(A'+B+C+D)(A'+B+C'+D)(A'+B'+C+D)(A'+B'+C'+D)$

From truth table

Problem #3

(a) $F(x,y,z) = \Sigma(1,3,5) \rightarrow F(x,y,z) = \Pi(0,2,4,6,8)$

(b) $F(A,B,C,D) = \Pi(3,5,8,11) \rightarrow F(A,B,C,D) = \Sigma(0,1,2,4,6,7,9,10,12,13,14,15)$

Problem #4

(a) $xy + x(y+z) + y(y+z)$

$= xy + xy + xz + y^2 + y^2 + yz$

$= xy + xz + y^2 + yz$

$= y + xz$

(b) $x'y'(x+y)(x+z)$

$= x'y'$

(c) $x + y'x'$

$= (x + x')x' = x' = 1$

* (d) $(x+y+z)(x'+y'+z)$ if $z=1 \rightarrow x+y$

$= z(x+y) + z'(x'+y') = x+y$ if $z=0 \rightarrow x'+y'$

problem #5

$$(a) \bar{A} \cdot (A+B) = \overset{0}{\bar{A}A} + \bar{A}B = \bar{A}B \checkmark$$

$$(b) (\bar{A} \cdot B) + (A \cdot B) = B(\overset{1}{\bar{A}+A}) = B \checkmark$$

$$(c) (A+B)(A+C) = A + (BC) \checkmark \quad \text{distribution}$$

$$(d) (A \cdot B) + (\bar{A} \cdot B) + (A \cdot \bar{B})$$

$$\begin{aligned} & \underbrace{B(A + \bar{A})}_{\overset{1}{B}} = B \\ & = \bar{A}B + (A\bar{B}) \\ & = (B+A)(B+\bar{B}) \overset{1}{=} \\ & = A+B \checkmark \end{aligned}$$

$$\begin{aligned} (e) & (A+C)(\underline{A\bar{D} + A\bar{D}}) + AC + C \\ & = (A+C)A + AC + C \\ & = \underline{A + AC} + C \quad \text{Absorption} \\ & = A + C \checkmark \end{aligned}$$

$$(f) A'(CA+B) + (B + \overset{A}{A \cdot A})(CA+B')$$

$$\begin{aligned} & \overset{0}{=} \bar{A}'\bar{A} + \bar{A}B + (B+A)(CA+B') \\ & = \bar{A}'B + A(\bar{B}\bar{B}') \\ & = \bar{A}'B + A \\ & = \underset{1}{(A+\bar{A})}(CA+B) = A+B \checkmark \end{aligned}$$